Financial Literacy: Budgeting and Saving - RAG System Report

1. Introduction

This report presents the design and development of a Retrieval-Augmented Generation (RAG) system built for the Financial Literacy domain, focusing on budgeting, savings, and investment-related knowledge. A total of 100+ financial documents of 0.7 Gb were used, covering subdomains such as stocks, gold, taxation, and personal finance. The system was designed to help retrieve accurate financial insights efficiently by combining retrieval-based and semantic reranking methods.

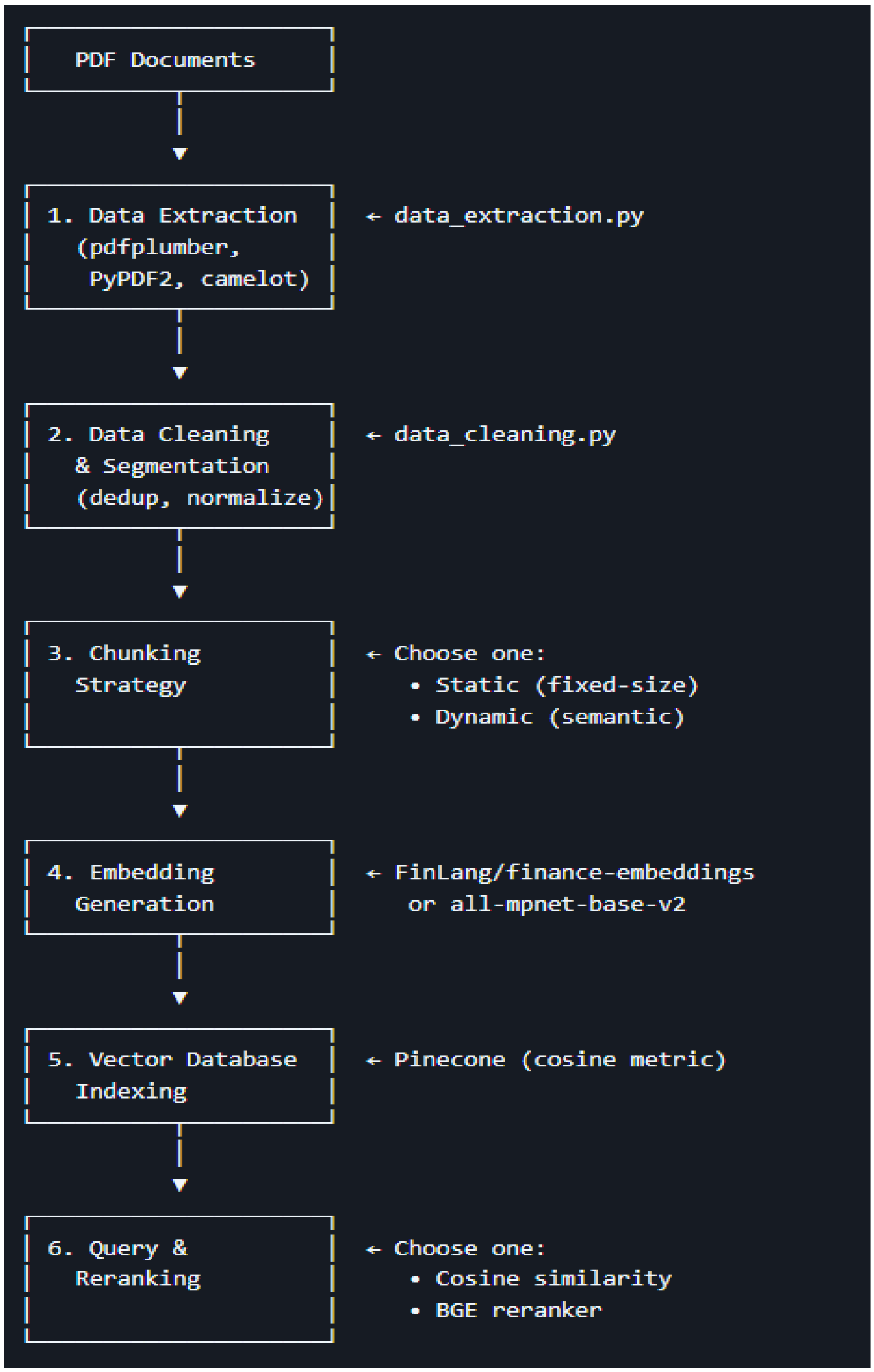
2. Dataset

This study utilizes a comprehensive dataset comprising over 100 PDF documents focused on finance, banking, and financial literacy with particular emphasis on the Indian financial landscape. The dataset encompasses a diverse range of materials including foundational financial education resources, regulatory certification workbooks from organizations such as SEBI and NISM, academic research on financial behavior and literacy, industry reports on banking and securities markets, investment guides covering stocks, real estate, and gold, and core financial and accounting textbooks. The collection covers critical topics such as personal budgeting, mutual funds, derivatives trading, monetary policy, taxation, risk management, and anti-money laundering provisions, making it a valuable resource for comprehensive analysis of financial education, investment patterns, and market dynamics in India. By combining regulatory materials, academic research, and industry insights, this dataset provides a multifaceted perspective on financial systems, investment opportunities, and financial literacy initiatives in the Indian context.

Dataset: https://huggingface.co/datasets/Karan007/financial-literacy-lma

3. Methodology

The complete RAG pipeline consists of four primary stages: data extraction, data cleaning and preprocessing, multi-granularity embedding generation with indexing, and query retrieval with reranking. Each stage is described below in detail.



3.1 Data Cleaning and Segmentation

The data cleaning phase was implemented using the script 'data_cleaning.py'. Each document underwent cleaning to remove duplicate lines, non-breaking spaces, and formatting artifacts. The text was normalized to lowercase and tokenized using NLTK's sentence tokenizer. The processed data was stored as 'knowledge_base_cleaned_segmented.jsonl'.

3.2 Multi-Granularity Embedding and Indexing

The 'data_embedding.py' script was responsible for embedding generation and vector indexing. We used the **'FinLang/finance-embeddings-investopedia'** model from Sentence Transformers for domain-specific embeddings. Each cleaned text segment was split into multiple granularities of 128, 512, and 2048 tokens, with a 50-token overlap to preserve context continuity. These embeddings (dimension 768) were stored in a Pinecone vector database using cosine similarity as the distance metric. Metadata such as document ID, page number, and granularity level were retained for interpretability.

3.3 Query Retrieval and Reranking

The 'reranking.py' module handles user queries and implements semantic retrieval. When a query is issued, its embedding is computed using the same 'FinLang' model. An initial candidate set (approximately 7× the top-k) is fetched from the Pinecone index. To improve relevance, the system employs a reranking mechanism using **'BAAI/bge-reranker-v2-m3'** from the FlagEmbedding library, which reorders the results based on semantic similarity. This ensures that the most contextually relevant chunks are prioritized.

3.4 Justification of Chunking and Reranking Strategy

3.4.1 Chunking Strategy

The main function responsible for chunking is segment_text_hybrid(text). It operates by first structuring the text based on its natural document hierarchy and then applying semantic grouping, followed by fixed-size token-based splitting.

3.4.1.1 Fixed Chunking strategy using Independent multi granularity chunks

Fixed chunking strategy divides text into equal-sized segments based on a set number of tokens, words, or characters, without considering sentence meaning. It ensures uniform chunk sizes, making it simple and efficient for embedding and indexing. However, it may split sentences or lose context, so small overlaps are often added between chunks to preserve continuity.

3.4.1.2 Document/Paragraph Splitting and Semantic chunking

The raw text is first split into paragraphs using newline characters (\n).

This ensures that the chunking process respects the original document structure, preventing chunks from spanning across distinct paragraphs that might discuss different topics.

Each paragraph is further divided into sentences using nltk.tokenize.sent_tokenize.

The semantic_chunk function then groups consecutive sentences. A new "semantic chunk" is started if the cosine similarity between the embeddings of the current sentence and the previous one drops below a defined SEMANTIC_THRESHOLD (0.55).

This aims to keep sentences that are topically cohesive together, as a drop in semantic similarity often indicates a topic shift.The final, largest step is to take the semantically-grouped chunks and split them into smaller, fixed-size chunks based on token count and store in parent-child relationship.

A crucial feature is the overlap mechanism: consecutive child chunks overlap by OVERLAP_TOKENS (50 tokens).

Purpose of Overlap: This preserves context that might otherwise be cut off at the boundaries, ensuring that key information or sentence continuity isn't lost between chunks.

In this project, a parent-child chunking strategy was used with chunk sizes of 128, 512, and 2048 tokens, maintaining a 50-token overlap between consecutive chunks. This approach was implemented to handle the heterogeneous nature of financial documents, which range from short definitions to long regulatory explanations.

3.4.2 Reranking Strategy

3.4.2.1 Cosine Similarity

Cosine similarity is a measure used in vector databases like Pinecone to determine how similar two embedding vectors are in terms of their direction rather than magnitude. It calculates the cosine of the angle between two vectors. For example, the embeddings of "Emergency fund helps in crisis" and "Savings for unexpected events" would have a cosine similarity close to 1, indicating they represent similar financial concepts, whereas unrelated sentences like "Invest in long-term stocks" would have a much lower similarity score.

3.4.2.2 After initial retrieval, the top candidates are reranked using the **BGE Reranker (BAAI/bge-reranker-v2-m3)** from the FlagEmbedding library.

Why Use a Reranker?

- The first retrieval (using Pinecone) is purely **vector-similarity-based**, which captures surface-level semantic closeness but can miss nuanced contextual relationships.
- The reranker computes a **contextual relevance score** between the query and each retrieved chunk — ensuring results that *semantically answer* the question are prioritized.

4. Example Retrieval Result

Experiment 1 (Fixed chunking) : An example query was tested to evaluate the RAG system's performance. The query used was: 'What is an emergency fund?'. Below are the top retrieved and reranked results:

Rank	Source	Initial Score	Excerpt
1	NISM Series X-A Investment Adviser Level 1.pdf	0.5315	The fund should be adequate to meet the expenses for six months...

2	Financial-Literacy-Basics-for-Adults.pdf	0.4928	Step 4: Emergency fund building and maintaining an emergency fund...
3	NISM Series X-A Investment Adviser Level 1.pdf	0.4868	If the emergency fund is used then efforts must be made to replenish it...

Experiment 2 (Dynamic chunking): An example query was tested to evaluate the RAG system's performance. The query used was: *'What is an emergency fund?'*. Below are the top retrieved and reranked results:

Rank	Source	Initial Score	Excerpt
1	book3.pdf	0.4239	an etf is a basket of securities that can trade like stocks on a stock exchange.
2	Deutsche-Bank-Namaste-India-2025.pdf	0.4072	'specified fund' is defined in the explanation to section 10(4d) of the it act.
3	book3.pdf	0.2976	this market is referred to as the market for loanable funds. in the market for loanable funds, the suppliers of funds are economic entities that currently..
4	Deutsche-Bank-Namaste-India-2025.pdf	0.2971	category i aif: invests in start-up or early-stage ventures or social ventures or smes or infrastructure or other sectors or areas which the government or regulators consider as socially or economically desirable and shall include venture capital funds, sme funds, social impact funds, …
5	book3.pdf	0.2756	federal reserve (fed) targets the equilibrium interest rate on federal funds as one of its most important monetary policy tools. the federal funds market traditionally consists of the overnight …

5. Summary

This RAG system successfully integrates domain-specific financial embeddings and semantic reranking to improve retrieval accuracy across multiple financial topics such as budgeting, emergency funds, and investments. The use of Pinecone for scalable vector storage and the BGE reranker for enhanced contextual ranking has significantly improved retrieval precision while minimizing redundancy.